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PCB Manufacturing Terms Glossary

These terms are those used by PCB manufacturers. Some terms have a standard definition, but when used as a manufacturing term, take on a different meaning. These glossaries are shown in their entirety from the contributo

Term	Definition
Activating	A treatment that renders nonconductive material receptive to electroless deposition. Also called seeding, catalyzing, and sensitizing.
Additive Process	Any process in printed circuit board manufacturing where the circuit pattern is produced by the addition of metal.
Analytical Services Lab	Performs various tests such as plating thickness, inner layer connections to hole walls, photos or x-rays of circuit boards when required.
Annular Ring	The conductive material surrounding a hole.
Artwork Master	An accurately scaled (usually 1:1) pattern, which is used to produce the production master.
Aspect Ratio	The ratio of the circuit board thickness to the smallest drilled hole diameter.
B-Stage Material	Sheet material (fiberglass cloth) impregnated with a resin cured to an intermediate stage (B-stage resin). Pre-preg is the preferred term.
Back Planes and Panels	Interconnection panels onto which printed circuits, other panels, or integrated circuit packages can be plugged or mounted. Typical thickness is 0.125"–0.300."
Barrel	The cylinder formed by plating a drilled hole.
Base Copper	Copper foil provided in sheet form or clad to one or both sides of piece of laminate used as either internal or external layers of a circuit board.
Base Laminate	The dielectric material upon which the conductive pattern may be formed. The base material may be rigid or flexible.
Base Material	See Base Laminate.
Bed-of-Nails Technique	A method of testing printed circuit boards that employs a test figure mounting an array of contact pins configured so as to engage plated thru-holes on the board.

Term	Definition
Bleeding	A condition in which a plated hole discharges process material or solution from crevices or voids.
Blind Via Hole	A plated-through hole connecting an outer layer to one or more internal conductor layers of a multilayer_ printed board but not extending fully through all of the layers of base material of the board.
Blister	A localized swelling and separation between any of the layers of a laminated base material, or between base material and conductive foil. It is a form of delamination.
Blow Hole	A solder joint void caused by out-gassing of process solutions during thermal cycling.
Bond Strength	The force per unit area required to separate two adjacent layers of a board when applied perpendicular to the board surface. See Peel Strength.
Breakdown Voltage	The voltage at which an insulator or dielectric ruptures, or at which ionization and conduction take place in a gas or vapor.
Bridging, Electrical	The formation of a conductive path between two insulated conductors such as adjacent traces on a circuit board.
Buildability Meeting	Team meeting to review customer designs against manufacturing process capabilities. Used to identify possible failure modes prior to fabrication.
C-Stage	The condition of a resin polymer when it is in the fully cured, cross-linked solid state, with high molecular weight.
Center-to-Center Spacing	The nominal distance between the centers of adjacent features or traces on any layer of a printed circuit board. Also known as "pitch."
Chamfer	A corner that has been rounded to eliminate an otherwise sharp edge.
Characteristic Impedance	A compound measurement of the resistance, inductance, conductance and capacitance of a transmission line expressed in ohms. In printed wiring its value depends on the width and thickness of the conductor, the distance from the conductor to ground plane(s), and the dielectric constant of the insulating media.
Chase	The aluminum frame used in screening inks onto the board surface.
Circuitry Layer	The layer of a PCB containing conductors, including ground and voltage planes.
Clad or Cladding	A thin layer or sheet of copper foil, which is bonded to a laminate core to form the base material for printed circuits. See Base Copper.
Clearance Hole	A hole in the conductive pattern larger than, but concentric with, a hole in the printed board base material.
Coefficient of Expansion, Thermal	The fractional change in dimension of a material for a unit change in temperature.
Component Hole	A hole used for the attachment and electrical connection of component terminations, including pins and wires, to the printed circuit board.
Component Side	That side of the printed circuit board on which most of the components will be mounted.
Conductive Pattern	The configuration or design of the conductive material on the base laminate through which electrical energy passes. Includes conductors, lands, and through connections.
Conductor	A thin conductive area on a PCB surface or internal layer usually composed of lands (to which component leads are connected) and paths (traces).
Conductor Base Width	The conductor width at the plane of the surface of the base material. See Conductor Width.
Conductor-to-Hole Spacing	The distance between the edge of a conductor and the edge of hole.

Term	Definition
Conductor Thickness	The thickness of the trace or land including all metallic coatings.
Conductor Width	The observable width of the pertinent conductor at any point chosen at random on the printed circuit board.
Contaminant	An impurity or foreign substance whose presence on printed wiring assemblies could electrically, chemically, or galvanically corrode the system.
Continuity	An uninterrupted flow of electrical current in a circuit.
Coordinate Tolerancing	A method of tolerancing hole locations in which the tolerance is applied directly to linear and angular dimensions, usually forming a rectangular area of allowable variation. Also see Positional Limitation Tolerancing and True Position Tolerance.
Copper Foil	See Base Copper and Clad or Cladding.
Core Group	Daily operational meeting held on each shift to communicate current status of plant in terms of producing and delivering high quality, cost-effective circuit boards to customers on time. The group identifies problems that may prevent reaching daily/weekly goals and plans resolution of those problems.
Cosmetic Defect	A defect such as a slight change in its usual color that doesn't affect a board's functionality.
Cover Lay, Cover Layer, Cover Coat	Outer layer(s) of insulating material applied over the conductive pattern on the surface of a printed circuit board.
Crazing	A condition existing in the base material in the form of connected white spots or "crosses" on or below the surface of the base material, reflecting the separation of fibers in the glass cloth and resin material.
Current Carrying Capacity	The maximum current which can be carried continuously, under specified conditions, by a conductor without causing degradation of electrical or mechanical properties of the printed circuit board.
Datum Reference	A defined point, line, or plane used to locate the pattern or layer for manufacturing, inspection, or for both purposes.
Deburring	Process of removing traces of base copper material that remain around holes after board drilling.
Defect	Any deviation from the normally accepted characteristics of a product or component. Also see Major Defect and Minor Defect.
Definition	The accuracy of pattern edges in a printed circuit relative to the master pattern.
Delamination	A separation between any of the layers of a base material or between the laminate and the conductive foil, or both.
Desmear	Removal of epoxy smear (melted resin) and drilling debris from a hole wall.
Develop	An imaging operation in which unpolymerized (unexposed) photo-resist is dissolved or washed away to produce a copper board with a photo-resist pattern for etching or plating.
Dewetting	A condition that occurs when molten solder has coated a surface and then recedes, leaving irregularly shaped globules of solder separated by areas covered with a thin solder film; base metal is not exposed.
Dielectric	An insulating medium, which occupies the region between two or more conductors.
Dielectric Constant	The ratio of permittivity of the material to that of a vacuum (referred to as relative permittivity).
Digitizing	Any method of converting feature locations on a flat plane to digital representation in x-y coordinates.

Term	Definition
Dimensional Stability	A measure of dimensional change caused by factors such as temperature, humidity, chemical treatment, age or stress; usually expressed as units/unit.
Dimensioned Hole	A hole in a printed circuit board where the means of determining location is x-y coordinate values not necessarily coinciding with the stated grid.
Double-Sided Board	A circuit board with conductive patterns on both sides.
Drills, Circuit Board	Solid carbide cutting tools with four facet points and two helical flutes designed specifically for the fast removal of chips in extremely abrasive materials.
Dry Film Resists	Coating material in the form of laminated photosensitive sheets specifically designed for use in the manufacture of printed circuit boards and chemically machined parts. They are resistant to various electroplating and etching processes.
Edge Bevel	A bevel operation performed on edge connectors to improve their wear and ease of installation.
Edge-Board Connector	A connector designed specifically for making removable and reliable interconnection between the edge board contacts on the edge of a printed board and external wiring.
Edge Dip Solderability Test	A solderability test performed by taking a specially prepared specimen, fluxing it with a nonactivate_d rosin flux, and then immersing it into a pot of molten solder at a predetermined rate of immersion for a predetermined dwell time, and then withdrawing it at a predetermined rate.
Electroless Plating/Electroless Deposition	The disposition of metal from an auto catalytic plating solution without application of electrical current. Short for "electrodeless." This process is required to plate the nonconductive hole walls in order that they may be subsequently electroplated. Also called "PTH."
Electroplating	(1) The electrodeposition of a metal coating on a conductive object. The object to be plated is placed in an electrolyte and connected to one terminal of a DC voltage source. The metal to be deposited is similarly immersed and connected to the other terminal. Ions of the metal provide transfer to metal as they make up the current flow between the electrodes. (2) The electrolytic process used to deposit a metal on a desired object by placing the object at one electrical polarity and passing a current through a chemical solution to another electrode. The metal is plated from either the solution or the other electrode.
Entrapment	The damaging admission and trapping of air, flux, and/or fumes; it is caused by contamination and plating.
Epoxy Smear	Epoxy resin that has been deposited onto the surface or edges of the conductive inner layer pattern during drilling. Also called Resin Smear.
Etch	Chemical removal of metal (copper) to achieve a desired circuit pattern.
Etch Factor	The ratio of the depth of etch (conductor thickness) to the amount of lateral etch (undercut).
Etchback	The controlled removal of all components of base material (glass and resin) by a chemical process on the sidewall of holes in order to expose additional internal conductor areas.
Fiducial	Etched features or drilled hole used for optical alignment during assembly operations.
Film Artwork	A positive or negative piece of film containing a circuit, solder mask, or nomenclature pattern.
First Article	A sample part or assembly manufactured prior to the start of production for the purpose of assuring that the manufacturer is capable of manufacturing a product that will meet specified requirements.

Term	Definition
Fixture	A device that enables interfacing a printed circuit board with a spring-contact probe test pattern.
Flat	A standard size sheet of laminate material, which is processed into one or more circuit boards.
Flux	A substance used to promote or facilitate fusion such as a material used to remove oxides from surfaces to be joined by soldering or welding.
Fused Coating	A metallic coating (usually tin or solder alloy) that has been melted and solidified, forming a metallurgical bond to the base material.
Glass Transition Temperature	The temperature at which an amorphous polymer (or the amorphous regions in a partially crystalline polymer) changes from a hard and relatively brittle condition to a viscous or rubbery condition. When this transition occurs, many physical properties undergo significant changes. Some of those properties are hardness, brittleness, coefficient of thermal expansion, and specific heat.
Grid	An orthogonal network of two sets of parallel, equidistant lines used for locating points on a printed circuit board.
Ground Plane	A conductor layer, or portion of a conductor layer, used as a common reference point for circuit returns, shielding, or heat sinking.
Haloing	Mechanically induced fracturing delamination on or below the surface of the base material; it is usually exhibited by a light area around holes, or other machines areas, or both.
Hole Breakout	A condition in which a hole is not completely surrounded by the land.
Hole Density	The quantity of holes in a printed circuit board per unit area.
Hole Void	A void in the metallic deposit of a plated-through hole exposing the base material.
Image	That portion on artwork masters, working tools, silk screens, or photo masks that would be considered the photographic image. Also would include images created with photo-resists or silk-screening techniques. Generally, "one image" refers to a single circuit board image; thus there may be several images per flat.
Ink	Common term for screen resist.
Inner layer	Any layer that will be pressed on the inside of a multilayer_ board.
Inspection Overlay	A positive or negative transparency made from the production master and used as an inspection aid.
Insulation Resistance	The electrical resistance of the insulating material (determined under specified conditions) as measured between any pair of contacts or conductors.
Jumper Wire	An electrical connection formed by wire between two points on a printed board added after the intended conductive pattern is formed.
Kerf	A widening of the rout path as may be called out on the blueprint. Allows extra space for hardware to be attached to the board.
Keying Slot	A slot in a printed circuit board that polarizes it, thereby permitting it to be plugged into its mating receptacle with pins properly aligned, but preventing it from being reversed or plugged into any other receptacle.
Laminate	A product made by bonding together two or more layers of material.
Laminate Thickness	Thickness of the base material, not including metal-clad, prior to any processing. Applies to single- or double-sided material.
Laminate Void	Absence of laminate material in an area that normally contains laminate material.
Laminating Presses, Multilayer_	Equipment that applies both pressure and heat to laminate and pre-preg to make multilayer_ boards.
Lamination	The process of preparing a laminate.

Term	Definition
Land	A portion of a conductive pattern usually, but not exclusively, used for the connection and/or attachment of components. Also called Pad.
Landless Hole	A plated-through hole without land(s). Also referred to as padless plated holes.
Layer-to-Layer Spacing	The thickness of dielectric material between adjacent layers or conductive circuitry in a multilayer_ printed circuit board.
Lay-Up	(1) The technique of registering and stacking layers of materials (laminate and pre-preg) for a multilayer_ board in preparation for the laminating cycle. (2) The laying out of repeat images on film to create multiple groups of circuit boards. (3) The laying out of multiple layers in preparation for multilayer_ lamination.
Major Defect	A defect that could result in a failure or significantly reduces the usability of the part for its intended purpose.
Mask	A material applied to enable selective etching, plating, or the application of solder to a printed circuit board.
Measling	Condition existing in the base laminate in the form of discrete white spots or "crosses" below the surface of the base laminate, reflecting a separation of fibers in the glass cloth at the weave intersection.
Metlab	Short for Metallurgical Laboratory. (1) Refers to the process(es) of inspecting internal board quality characteristics through the use of microsections. (2) Used interchangeably with microsection.
Microsectioning	The preparation of a specimen for the microscopic examination of the material to be examined, usually by cutting out a cross section, followed by encapsulation, polishing, etching, staining, and so on.
Microvia	A via used to make connection between two adjacent layers, typically less than 6 mils in diameter. May be formed by laser ablation, plasma etching, or photo processing.
Mil	One-thousandth of an inch (0.001").
Minimum Annular Ring	The minimum metal width, at the narrowest point, between the circumference of the hole and the outer circumference of the land. This measurement is made to the drilled hole on internal layers of multilayer_ printed circuit boards and to the edge of the plating on outside layers of multilayer_ boards and double-sided boards.
Minimum Electrical Spacing	The minimum allowable distance between adjacent conductors that is sufficient to prevent dielectric breakdown, corona, or both, between the semiconductors at any given voltage and altitude.
Minor Defect	A defect that is not likely to reduce the usability of the unit for its intended purpose. It may be a departure from established standards having no significant bearing on the effective use or operation of the unit.
Misregistration	The lack of dimensional conformity between successively produced features or patterns.
Multilayer_ Circuit Board	The general term for completely processed printed circuit configurations consisting of alternate layers of conductive patterns and insulating materials bonded together in more than two layers.
Nail Heading	The flared condition of copper on the inner conductor layers of a multilayer_ board caused by hole drilling.
Negative	An artwork master or production master in which the intended conductive pattern is transparent to light, and the areas to be free from conductive material are opaque.
Nonfunctional Land	A land on internal or external layers, not connected to the conductive pattern on its layer.

Term	Definition
October Project	Industry consortium dedicated to waste reduction in circuit board manufacturing.
Outer Layer	A conductive layer that lies on the outside of a flat.
Outgassing	Deaeration or other gaseous emission from a printed circuit board when exposed to the soldering operation.
Overhang	Increase in printed circuit conductor width caused by plating build-up or by undercutting during etching.
Oxide	A chemical treatment to inner layers prior to lamination, for the purpose of increasing the roughness of clad copper to improve laminate bond strength.
Pad	The portion of the conductive pattern on printed circuits designated for the mounting or attachment of components. Also called Land.
Panel	The square or rectangular base material containing one or more circuit patterns that passes successively through the production sequence and from which printed circuit boards are extracted. See Back Planes and Panels.
Panel Plating	The electrolytic plating of the entire surface of a panel (including holes).
Part	Individual items that are not normally subject to disassembly without destruction.
Pattern	The configuration of conductive and nonconductive materials on a panel or printed board. Also the circuit configuration on related tools, drawings, and masters.
Pattern Plating	Selective electrolytic plating of a conductive pattern.
Peel Strength	The force required to peel the conductor or foil from the base material.
Permittivity	Measure of the ability of a material to store electrical energy when exposed to an electrical field.
Photomask	A silver halide or diazo image on a transparent substrate that is used to either block or pass light.
Photo-resist	A light-sensitive material that is used to establish an image by exposure to light and chemical development.
Photo plotter	A high-accuracy (>0.002 inch) flatbed plotter with a programmable, photo image projector assembly. It is most often used to produce actual-size master patterns for printed circuit artwork directly on dimensionally stable, high-contrast photographic film.
Pilot Order	First production order going through process.
Pinhole	A minute hole through a layer or pattern.
Pitch	The nominal distance between the centers of adjacent features or traces on any layer of a printed circuit board. Also known as "center-to-center spacing."
Plated-Through Hole (PTH)	A hole in a circuit board that that been plated with metal (usually copper) on its sides to provide electrical connections between conductive patterns layers of a printed circuit board.
Plating	Chemical or electromechanical deposition of metal on a pattern.
Plating Resists	Material that, when deposited on conductive areas, prevents the plating of the covered areas. Resists are available both as screened-on materials and as dry-film photopolymer resists.
Plating Void	The absence of a plating metal from a specified plating area.
Plotting	The mechanical converting of x-y positional information into a visual pattern, such as artwork.
Polymide Resins	High-temperature thermoplastics used with glass to produce printed circuit laminates for multilayer_ and other circuit applications requiring high-temperature performance.

Term	Definition
Polymerize	To unite chemically two or more monomers or polymers to form a molecule with a higher molecular weight.
Positional Limitation Tolerancing	Defines a zone within which the axis or center plane of a feature is permitted to vary from true (theoretically exact) position.
Pre-clean	Cleaning steps taken prior to an operation to ensure success of the operation.
Pre-preg	Sheet material consisting of the base material impregnated with a synthetic resin, such as epoxy or Polymide, partially cured to the B-stage (an intermediate stage). Short for pre-impregnated. See also B-stage.
Press-Fit Contact	An electrical contact that can be pressed into a hole in an insulator, printed board (with or without plated-through holes), or a metal plate.
Printed Circuit	A conductive pattern of printed components and circuits attached to a common base.
Printed Circuit Board (PCB)	The general term for a printed or etched circuit board. It includes single, double, or multiple layer boards, both rigid and flexible.
Printed Wiring Board	A part manufactured from rigid base material upon which completely processed printed wiring has been formed.
Production Master	A 1 : 1 scale pattern that is used to produce one or more printed boards (rigid or flexible) within the accuracy specified on the master drawing. (1) Single-Image Product Master: A production master used in the process of making a single printed circuit board. (2) Multiple-Image Production Master: A production master used in the process of making two or more printed circuit boards simultaneously.
Reflowing	The melting of an electrodeposit followed by solidification. The surface has the appearance and physical characteristics of being hot-dipped.
Registration	The degree of conformity of the position of a pattern, or a portion thereof, with its intended position or with that of any other conductor layer of a board.
Residue	An undesirable substance remaining on a substrate after a process step.
Resin Smear	Resin transferred from the base material onto the surface or edge of the conductive pattern normally caused by drilling. Sometimes called epoxy smear.
Resin-Starved Area	A region in a printed circuit board that has an insufficient amount of resin to wet out the reinforcement completely evidenced by low gloss, dry spots, or exposed fibers.
Resist	Coating material used to mask or to protect selected areas of a pattern from the action of an etchant, solder, or plating. Also see Dry-Film Resists, Plating Resists, and Solder Resists.
Resistivity	The ability of a material to resist the passage of electrical current through it.
Reverse Image	The resist pattern on a printed circuit board enabling the exposure of conductive areas for subsequent plating.
Rework	Reprocessing that makes articles conform to specifications.
Robber	An exposed area generally attached to a rack used in electroplating, usually to provide a more uniform current density on plated parts. Thieves are intended to absorb the unevenly distributed current on parts, thereby assuring that the parts will receive a uniform electroplated coating.
Schematic Diagram	A drawing that shows, by means of graphic symbols, the electrical connections, components, and functions of an electronic circuit.
Screen	A cloth material (usually polyester or stainless steel for circuit boards) coated with a pattern that determines the flow and location of coatings forced through its openings.

Term	Definition
Screen Printing	A process for transferring an image to a surface by forcing suitable media through a stencil screen with a squeegee. Also called silk screening.
Selective Plate	A process for plating unique features with a different metal than the remaining features will have. Created by imaging, exposing, and plating selected area and then repeating the process for the remainder of the board.
Shadowing	A condition occurring during etchback in which the dielectric material, in contact with the foil, is incompletely removed although acceptable etchback may have been achieved elsewhere.
Silk Screening	See Screen Printing.
Single-Sided Board	Circuit board with conductors on only one side and no plated-through holes.
Solder Leveling	The process of dipping printed circuit boards into hot solder and leveling with hot air.
Solder Mask	A coating applied to a circuit board to prevent solder from flowing onto any areas where it's not desired or from bridging across closely spaced conductors.
Solder Masking Coating	Nonpreferred term for resist.
Solder Resists	Coatings that mask and insulate portions of a circuit pattern where solder is not desired.
Solderability Testing	The evaluation of a metal to determine its ability to be wetted by solder.
Squeegee	The tool used in silk screening that forces the resist or ink through the mesh.
Starvation, Resin	A deficiency of resin in base material that is apparent after lamination by the presence of weave texture, low gloss, or dry spots.
Step-and-Repeat	A method by which successive exposures of a single image are made to produce a multiple-image production master.
Strip	The chemical removal of developed photoresist or plated metal.
Substrate	See Base Material.
Subtractive Process	A process in printed circuit manufacturing where the product is built by the subtraction of an already existing metallic coating. The opposite of additive processing.
Test Coupon	A sample or test pattern normally made outside the actual board pattern that is used for testing to verify certain quality parameters without destroying the actual board.
Thief	See Robber.
Tooling Holes	Two specified holes on a printed circuit board used to position the board in order to mount components accurately.
Traveler	A "recipe" for the manufacture of a board. It "travels" with each order from start to finish. The traveler identifies each order and gives instructions for each step in the process. It also provides information for traceability and history.
Two-Sided Board	See Double-Sided Board.
Underwriters Laboratory	Certifying agency for consumer electronics. See also Underwriters Symbol.
Underwriters Symbol	A logotype denoting that the product has been recognized by Underwriters Laboratory, Inc. (UL).
Via	A plated thru-hole that is used as an inner-layer connection but doesn't have component lead or other reinforcing material in it.
Void	The absence of substances in a localized area (e.g., air bubbles).

Term	Definition
Wave Soldering	A process wherein assembled printed boards are brought in contact with a continuously flowing and circulating mass of solder.
Zero Defects Sampling	A statistical based attribute sampling plan (C = O) where a given sample of parts is inspected and any defects found are cause for rejection of the entire lot.

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